

Table C.19: Summary of Bayesian Models

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
AIRSPACE OPACITY	ACCURACY	0.88	0.91	0.88	0.89	0.90	0.90	0.89	0.90	0.89	0.89	0.91	0.90
	AUC	0.89	0.96	0.91	0.92	0.96	0.96	0.94	0.96	0.93	0.93	0.96	0.96
CARDIOMEDIASTINAL ABNORM.	ACCURACY	0.90	0.92	0.90	0.91	0.93	0.92	0.92	0.92	0.91	0.91	0.93	0.93
	AUC	0.90	0.96	0.91	0.93	0.96	0.96	0.94	0.96	0.94	0.94	0.96	0.96
ABNORMAL	ACCURACY	0.88	0.91	0.88	0.89	0.91	0.91	0.90	0.91	0.90	0.90	0.92	0.91
	AUC	0.91	0.96	0.91	0.93	0.96	0.96	0.95	0.96	0.94	0.94	0.96	0.96
FOCAL s_A		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
OTHER s_A			✓			✓	✓		✓			✓	✓
FOCAL s_E				✓	✓	✓	✓			✓	✓	✓	✓
OTHER s_E					✓		✓				✓		✓
CLINICAL HISTORY s_E								✓	✓	✓	✓	✓	✓

Note: This table summarizes the models used to estimate the Bayesian benchmark. Each column corresponds to a random forest classification tree with varying signal structures. The rows Focal s^A , Other s^A , Focal s^E , Other s^E , and Clinical History s^E indicate what features are included in the tree. Focal s^A corresponds to the focal pathology's AI score, Other s^A corresponds to vector of AI scores for all pathologies, Focal (Other) s^E includes the radiologist's report without AI assistance on the focal pathology (all pathologies), and Clinical History s^E contains summaries of the patient's clinical history when available to the radiologist.

C.5.3 Model Selection on Additional Pathology Groups

Here, we present the model selection results for the remaining pre-registered pathology groups.

Table C.20: Model Selection: Top Level with AI

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
AUTOMATION BIAS (b)	0.29 (0.02)	0.35 (0.03)	0.34 (0.03)	0.31 (0.03)	0.35 (0.03)	0.35 (0.03)	0.19 (0.02)	0.34 (0.03)	0.27 (0.02)	0.25 (0.03)	0.34 (0.03)	0.35 (0.03)
OWN INFO BIAS (d)	1.10 (0.01)	1.06 (0.01)	1.08 (0.01)	1.08 (0.01)	1.06 (0.01)	1.05 (0.01)	1.07 (0.01)	1.06 (0.01)	1.06 (0.01)	1.07 (0.01)	1.06 (0.01)	1.05 (0.01)
CONSTANT	0.40 (0.03)	0.37 (0.04)	0.40 (0.04)	0.38 (0.04)	0.37 (0.04)	0.37 (0.04)	0.31 (0.03)	0.38 (0.04)	0.33 (0.03)	0.34 (0.04)	0.37 (0.04)	0.36 (0.04)
FOCAL s_A	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
OTHER s_A		✓			✓	✓		✓			✓	✓
FOCAL s_E			✓	✓	✓	✓			✓	✓	✓	✓
OTHER s_E				✓		✓				✓		✓
CLINICAL HISTORY s_E							✓	✓	✓	✓	✓	✓
J-STATISTIC	18.23	0.03	16.86	10.45	0.14	0.17	4.3	0.02	10.7	6.13	0.06	0.09
MMSC-BIC	-23.96	-8.41	-21.12	-10.64	-8.3	-8.27	-12.57	-8.42	-10.39	-10.75	-8.38	-8.35
NUMBER OF MOMENTS	13	5	12	8	5	5	7	5	8	7	5	5
Observations	11420	11420	11420	11420	11420	11420	11420	11420	11420	11420	11420	11420
R-SQUARED	0.50	0.52	0.49	0.49	0.52	0.52	0.46	0.51	0.48	0.46	0.52	0.52

Note: This table presents results of the model selection exercise as described in Table 3, including the full set of models that were included in the selection procedure.

Table C.21: Model Selection: Pooled with AI

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
AUTOMATION BIAS (b)	0.16 (0.01)	0.09 (0.01)	0.17 (0.02)	0.14 (0.02)	0.10 (0.01)	0.10 (0.01)	0.12 (0.01)	0.10 (0.01)	0.13 (0.02)	0.14 (0.02)	0.10 (0.01)	0.10 (0.01)
OWN INFO BIAS (d)	1.12 (0.01)	1.10 (0.01)	1.12 (0.01)	1.12 (0.01)	1.10 (0.01)	1.10 (0.01)	1.11 (0.01)	1.10 (0.01)	1.11 (0.01)	1.12 (0.01)	1.10 (0.01)	1.10 (0.01)
CONSTANT	0.37 (0.04)	0.31 (0.04)	0.36 (0.04)	0.37 (0.04)	0.31 (0.04)	0.31 (0.04)	0.34 (0.04)	0.31 (0.04)	0.34 (0.04)	0.36 (0.04)	0.31 (0.04)	0.31 (0.04)
FOCAL s_A	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
OTHER s_A		✓			✓	✓		✓			✓	✓
FOCAL s_E			✓	✓	✓	✓			✓	✓	✓	✓
OTHER s_E				✓		✓				✓		✓
CLINICAL HISTORY s_E							✓	✓	✓	✓	✓	✓
J-STATISTIC	12.52	3.85	10.53	4.81	2.03	2.71	4.64	3.84	3.98	5.2	2.44	2.4
MMSC-BIC	-12.79	-13.03	-10.57	-12.07	-14.85	-14.17	-12.24	-13.03	-12.9	-11.68	-14.43	-14.48
NUMBER OF MOMENTS	9	7	8	7	7	7	7	7	7	7	7	7
Observations	57100	57100	57100	57100	57100	57100	57100	57100	57100	57100	57100	57100
R-SQUARED	0.45	0.42	0.44	0.43	0.42	0.42	0.43	0.42	0.43	0.43	0.42	0.42

Note: This table presents results of the model selection exercise as described in Table 3, though this table only includes all pathologies with AI assistance.